

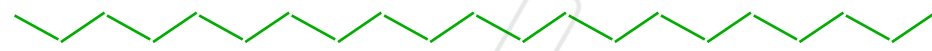
Number average molecular weight

$$M_1 = 100$$

$$N_1 = 1$$

$$M_2 = 10$$

$$N_2 = 1$$



number fraction : $n_i = \frac{\text{number of type } i}{\text{total number}}$

$$\rightarrow n_1 = 1/2 ; n_2 = 1/2$$

$$\bar{M}_n = \sum n_i M_i = \frac{1}{2} \cdot 100 + \frac{1}{2} \cdot 10 = 55$$

$$\bar{M}_n = \frac{\sum n_i M_i}{\sum n_i}$$

$$n_i = \frac{N_i}{\sum N} \quad \therefore n_1 = \frac{1}{2}, \quad n_2 = \frac{1}{2}$$

$$w_i = \frac{N_i M_i}{\sum N_i M_i} \quad \therefore w_1 = \frac{10}{11}, \quad w_2 = \frac{1}{11}$$

$$z_i = \frac{N_i M_i^2}{\sum N_i M_i^2} \quad \therefore z_1 = \frac{100}{101}, \quad z_2 = \frac{1}{101}$$

$$\therefore M_n = \sum n_i M_i = 55$$

$$M_w = \sum w_i M_i = \frac{1010}{11} = 91.82$$

$$M_z = \sum z_i M_i = \frac{10010}{101} = 99.12$$

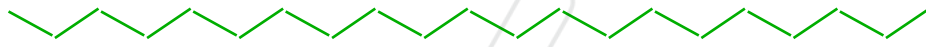
Weight average molecular weight

$$M_1 = 100$$

$$N_1 = 1$$

$$M_2 = 10$$

$$N_2 = 1$$



weight fraction : $w_i = \frac{\text{weight of type } i}{\text{total weight}}$

$$\rightarrow w_1 = 10/11 ; w_2 = 1/11$$

$$\bar{M}_w = \sum w_i M_i = \frac{10}{11} \cdot 100 + \frac{1}{11} \cdot 10 = 91.8$$

$$\bar{M}_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

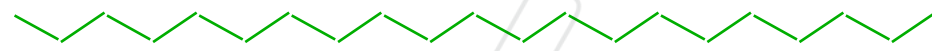
Z-average molecular weight

$$M_1 = 100$$

$$N_1 = 1$$

$$M_2 = 10$$

$$N_2 = 1$$



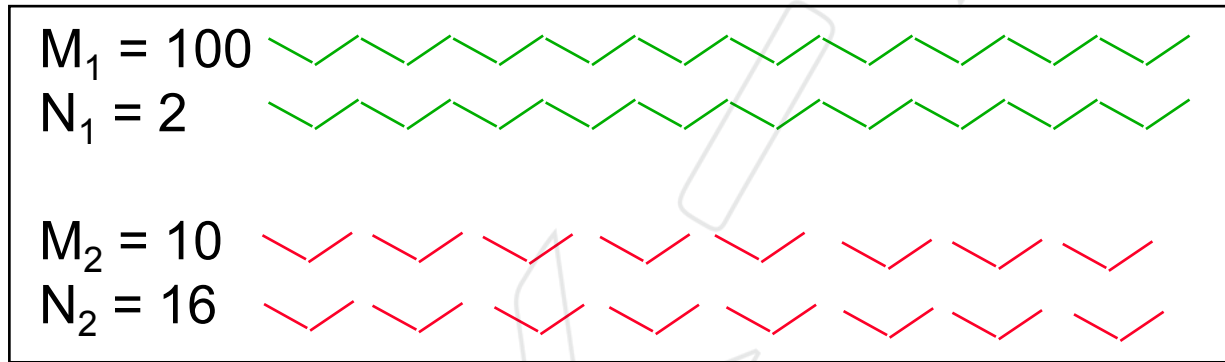
$$\text{Z-fraction: } z_i = \frac{N_i M_i^2}{\sum N_i M_i^2}$$

$$\rightarrow z_1 = 100/101 ; z_2 = 1/101$$

$$\bar{M}_z = \sum z_i M_i = \frac{100}{101} \cdot 100 + \frac{1}{101} \cdot 10 = 99.1$$

$$\bar{M}_z = \frac{\sum n_i M_i^3}{\sum n_i M_i^2}$$

Number average molecular weight



number fraction : $n_i = \frac{\text{number of type } i}{\text{total number}}$

$$\rightarrow n_1 = 2/18 ; n_2 = 16/18$$

$$\bar{M}_n = \sum n_i \cdot M_i = \frac{2}{18} \cdot 100 + \frac{16}{18} \cdot 10 = 20$$

$$\bar{M}_n = \frac{\sum n_i M_i}{\sum n_i}$$

$$\sum N_i = 18$$

$$\sum N_i M_i = 360$$

$$\sum N_i M_i^2 = 21600$$

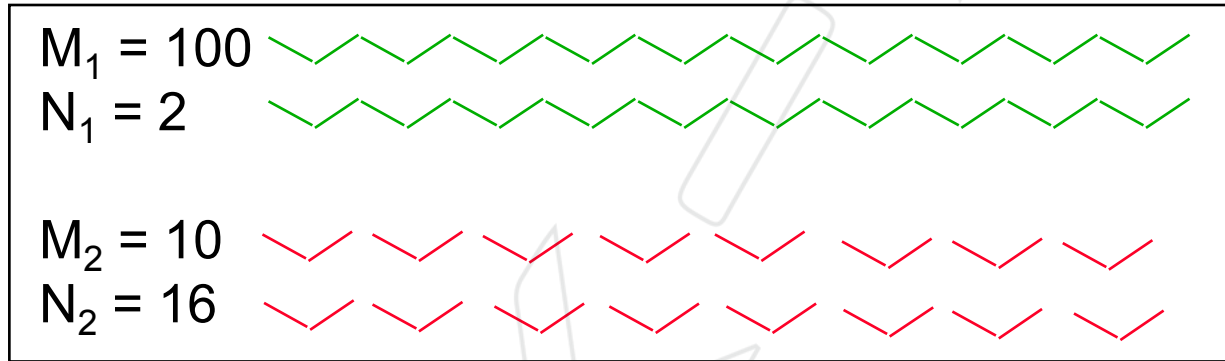
$$\sum N_i M_i^3 = 2016000$$

$$\therefore \bar{M}_n = \frac{\sum N_i M_i}{\sum N_i} = 20$$

$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = 60$$

$$\bar{M}_z = \frac{\sum N_i M_i^3}{\sum N_i M_i^2} = 93.3$$

Weight average molecular weight



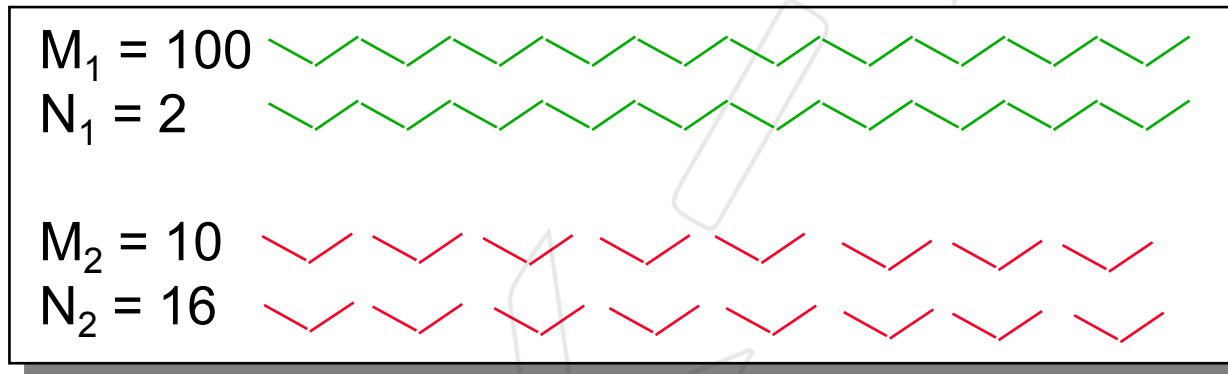
weight fraction : $w_i = \frac{\text{weight of type } i}{\text{total weight}}$

$$\rightarrow w_1 = 200/360 ; w_2 = 160/360$$

$$\bar{M}_w = \sum w_i \cdot M_i = \frac{200}{360} \cdot 100 + \frac{160}{360} \cdot 10 = 60$$

$$\bar{M}_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

Z-average molecular weight



$$\text{Z-fraction: } z_i = \frac{N_i M_i^2}{\sum N_i M_i^2}$$

$$\rightarrow z_1 = 200/216 ; z_2 = 16/216$$

$$\bar{M}_z = \sum z_i \cdot M_i = \frac{200}{216} \cdot 100 + \frac{16}{216} \cdot 10 = 93.3$$

$$\bar{M}_z = \frac{\sum n_i M_i^3}{\sum n_i M_i^2}$$

Example of averages calculations

9 molecules of $M = 30,000$ and 5 molecules of $M = 50,000$

- Number-average molar mass:

$$\overline{M}_n = \frac{\sum_{i=1}^{\infty} M_i N_i}{\sum_{i=1}^{\infty} N_i} = \frac{(9 \times 30,000) + (5 \times 50,000)}{(9 + 5)} = 37,000$$

- Weight-average molar mass:

$$\overline{M}_w = \frac{\sum_{i=1}^{\infty} w_i M_i}{\sum_{i=1}^{\infty} w_i} = \frac{\sum_{i=1}^{\infty} N_i M_i^2}{\sum_{i=1}^{\infty} N_i M_i} = \frac{9(30,000)^2 + 5(50,000)^2}{9(30,000) + 5(50,000)} = 40,000$$

- Z-average molar mass:

$$\overline{M}_z = \frac{\sum_{i=1}^{\infty} N_i M_i^3}{\sum_{i=1}^{\infty} N_i M_i^2} = \frac{\sum_{i=1}^{\infty} w_i M_i^2}{\sum_{i=1}^{\infty} w_i M_i} = \frac{9(30,000)^3 + 5(50,000)^3}{9(30,000)^2 + 5(50,000)^2} = 42,136$$

$$\sum N_i = 14$$

$$\sum N_i M_i = 520000$$

$$\sum N_i M_i^2 = 2.06 \times 10^{10}$$

$$\sum N_i M_i^3 = 8.68 \times 10^{14}$$

$$\therefore \bar{M}_n = \frac{\sum N_i M_i}{\sum N_i} = 37142.86$$

$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = 39615.38$$

$$\bar{M}_z = \frac{\sum N_i M_i^3}{\sum N_i M_i^2} = 42135.92$$